ENGS 3302 PATTERN RECOGNITION · WEEK GUIDE

Week 1: Introduction to Pattern Recognition

Week of 8 October 2026 · Videos ≈ 87 min · Lab video ≈ 45 min

Welcome! Start with the two short opening videos: why this course matters, and how it works week by week.

This week in one sentence. A machine recognises patterns by measuring *features* and drawing a *decision boundary* between the classes; good features and a boundary that works on new data are everything.

Learning outcome (T1): explain the parts of a pattern recognition system and the types of learning.

What to do, in this order

	Step	Notes	Time
☐	**Video 1.0** Teach a computer to see (motivation)	—	8 min
☐	**Video 1.0b** How this course works	—	14 min
☐	**Video 1.1** What is pattern recognition?	§1.1	10 min
☐	**Video 1.2** Sorting fish: choosing a feature (uses the interactive page *Feature Space Explorer*)	§1.2	20 min
☐	**Video 1.3** Feature space and generalization	§1.3–1.4	20 min
☐	**Video 1.4** Systems, design and learning	§1.5–1.7	15 min
☐	Read the Notes again, and do *Check yourself* at the end	all	40 min
☐	**Lab video** and `Lab_W01.ipynb`: Colab, NumPy, Pandas, Iris and handwritten digits	—	≈ 2 h
☐	`Assignment_W01.ipynb`: check-in, **no grade** **submit by Wed 14 Oct, 23:59**	—	30 min

Files for this week

- `W01_Notes.pdf` — **the study material. The exam covers what is in the Notes.**
- Slides used in the videos (for viewing only, not for study).
- `Lab_W01.ipynb`, `Assignment_W01.ipynb`.
- Interactive page: *Feature Space Explorer* (download the file and open it in your browser; it works offline).

Need help?

Ask on the course Discord server (invite link on Moodle) or by e-mail. Questions asked on Discord are answered for everyone. In-person discussion sessions: every two weeks for each section, Thursday 12:00–14:00 (female section in odd weeks, male section in even weeks). If you lose power or internet this week, download the Notes and the lab notebook as soon as you can: the Notes can be read offline, and Lab 1 shows how to run notebooks on your laptop without internet.